

years. Then, the Eq. $X(4)$ (equivalent $n = 11$ per year and $i = 8\%$; = JD 10001. Given a Salvage Value single value at the end of year 4) is

-428.88a.

(الإجابة الصحيحة) -1713.8b.

-583.49c.

per year, the Withdrawal $i = 10\%$, at a Simple Interest Rate JD 15002. For an original (first) amount of (Total) Amount after 2 years is

JD 300a.

JD 30 000b.

(الإجابة الصحيحة) JD 1800c.

years, then $n = 12$ per year and $i = 8\%$, if = 10003. A project with an Annual Income for the first 4 years the Equivalent Annual Value for the last 4 years is closest to:

(الإجابة الصحيحة) -1851a.

-540.3b.

-1000c.

per year compounded yearly is the same as 8%4. An interest rate of

per year (الإجابة الصحيحة) 8.29%a. an effective

per year 8%b. an effective

per month 8.29%c. an effective

$i = 8\%$ years. If $n = 9$ yearly over 100, increased by = JD 10005. A project with an annual operating cost is around: (Eq. $X(0)$) per year, then the equivalent single payment at time zero

2180.8a.

8427.6b.

(الإجابة الصحيحة) 7527.2c.

is $(P/A, i, n) \cdot (F/P, i, n) = (-?-)$ 6. The missing Factor in the equation

$(P/F, i, n)$ a.

(الإجابة الصحيحة) $(F/A, i, n)$ b.

$(A/F, i, n)$ c.

years. Then, the Eq. $X(2)$ (equivalent $n = 11$ per year and $i = 8\%$; = JD 10007. Given a Salvage Value single value at the end of year 2) is

-1469.9a.

-428.88b.

(الإجابة الصحيحة) -500.25c.

per year, the Interest $i = 10\%$, at a Simple Interest Rate JD 15008. For an original (first) amount of Amount (I) after 2 years is

JD 1800a.

(الإجابة الصحيحة) JD 300b.

JD 30 000c.

per year, the Interest $i = 10\%$, at a Simple Interest Rate JD 15009. For an original (first) amount of Amount (I) after 5 years is

JD 2250a.

JD 75 000b.

(الإجابة الصحيحة) JD 750c.

years. If $n = 7$ yearly over 100, increased by = JD 100010. A project with an annual operating cost is around: (Eq. $X(0)$) per year, then the equivalent single payment at time zero $i = 8\%$

(الإجابة الصحيحة) 5675.1a.

1402.4b.

per year $i = 8\%$. If $S.V. = 500$, and $a = 1300$ 11. For a project with an Extra Cost at the end of the 8th year years, then the Equivalent Single Value at the end of the 8th year $X(8)$ is $n = 12$ and

932.50a.

+932.5b.

(الإجابة الصحيحة) -1667.5c.

; Annual Operating cost = JD 700012. Given a Project with the following data: First (initial) cost years. To Calculate the Equivalent $n = 12$ per year, and $i = 8\%$; = JD 1500; Salvage Value = JD 1200 Single Value (Eq. X) at the end of year 4, we use

$7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$ a.

(الإجابة الصحيحة) $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 4) - 1500(P/F, 8\%, 8)$ b.

$7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$ c.

years. Then, the Eq. X(5) (equivalent $n = 11$ per year and $i = 8\%$; = **JD 1000**13. Given a Salvage Value single value at the end of year 5) is

-428.88a.

(الإجابة الصحيحة) -11586.8b.

per year $i = 8\%$. If $S.V. = 500$, and $a = 1300$ 14. For a project with an Extra Cost at the end of the 9th year years, then the Equivalent Single Value at the end of the 9th year X(9) is $n = 12$ and

+903.10a.

(الإجابة الصحيحة) -1696.9b.

-903.1c.

, Annual Operating cost = **JD 7000**15. Given a Project with the following data: First (initial) cost years. To Calculate the Equivalent $n = 12$ per year, and $i = 8\%$; = **JD 1500**; Salvage Value = **JD 1200** Single Value (Eq. X) at the end of year 7, we use

(الإجابة الصحيحة) $7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12)(F/P, 8\%, 7) - 1500(P/F, 8\%, 5)$ a.

$7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 5)$ b.

$7000(F/P, 8\%, 6) + 1200(P/A, 9\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 5)$ c.

per year, the Withdrawal $i = 10\%$, at a Simple Interest Rate **JD 1500**16. For an original (first) amount of (Total) Amount after 7 years is

JD 105000a.

(الإجابة الصحيحة) **JD 2550**b.

JD 1050c.

1. An interest rate of 10% per year compounded semiannually is the same as:

- a. an effective 10.25% per year.**
- b. an effective 5% per quarter compounded continuously.
- c. an effective 10% per month.

2. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 5 years is:

- a. JD 750**
- b. JD 75 000
- c. JD 2250

3. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 3 years is:

- a. JD 1950
- b. JD 450**
- c. JD 45 000

4. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 12 years is:

- a. JD 3300**
- b. JD 180 000
- c. JD 1800

5. An interest rate of 1.5% per month compounded continuously is the same as:

- a. an effective 1.511% per month.
- b. an effective 1.5% per month.
- c. an effective 18% per year.**

6. An interest rate of 12% per semiannually compounded monthly is the same as:

- a. an effective 26.8% per year.
- b. an effective 1.5% per month.
- c. an effective 12.16% per year.**

7. The missing Factor in the equation $(-?-)(F/P, i, n) = (F/A, i, n)$ is:

- a. $(P/A, i, n)$**
- b. $(F/A, i, n)$
- c. $(F/P, i, n)$

8. The missing factor in the equation $(A/G, i, n)(-?-) = (P/G, i, n)$ is:

- a. $(G/P, i, n)$
- b. $(A/P, i, n)$
- c. $(P/A, i, n)$**

9. The missing Factor in the equation $(P/A, i, n)(F/P, i, n) = (-?-)$ is:

- a. $(P/F, i, n)$
- b. $(F/A, i, n)$
- c. $(A/F, i, n)$**

10. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(1)$ (equivalent single value at the end of year 1) is:

- a. -1080
- b. -463.2
- c. -428.88**

11. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(8)$ (equivalent single value at the end of year 8) is:

- a. -793.83**
- b. -428.88
- c. -1259.7

12. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(10)$ (equivalent single value at the end of year 10) is:

- a. -428.88
- b. -1080
- c. -925.93

13. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 1, we use:

- a. $7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 11)$
- b. $7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$
- c. $7000 + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$

14. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 4, we use:

- a. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$
- b. $7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$
- c. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$

15. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 8, we use:

- a. $7000(F/P, 8\%, 8) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 4)$
- b. $7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12)(F/P, 8\%, 7) - 1500(P/F, 8\%, 4)$
- c. $7000(F/P, 8\%, 8) + 1200(P/A, 8\%, 12)(F/P, 8\%, 8) - 1500(P/F, 8\%, 4)$

16. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 10, we use:

- a. $7000(F/P, 8\%, 9) + 1200(P/A, 8\%, 12)(F/P, 8\%, 9) - 1500(P/F, 8\%, 2)$
- b. $7000(F/P, 8\%, 10) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 2)$
- c. $7000(F/P, 8\%, 10) + 1200(P/A, 8\%, 12)(F/P, 8\%, 10) - 1500(P/F, 8\%, 2)$

17. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 4$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

a. 2821.6

b. 244.5

c. 3777.1

18. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 8$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

a. 7527.2

b. 1780.6

c. 6608.7

19. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 9$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

a. 7527.2

b. 8427.6

c. 2180.8

20. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 11$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

a. 10165.4

b. 3026.5

c. 9307.6

مادة الاقتصاد الهندسي - امتحان النصف الفصل (Midterm)

أسئلة ومسابئلة محلولة بدقة

Q1: An interest rate of 10% per year compounded semiannually is the same as:

- a. an effective 10% per month.
- b. an effective 10.25% per year.
- c. an effective 5% per quarter compounded continuously.

Correct Answer: b. an effective 10.25% per year.

Q2: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 11$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

- a. 9307.6
- b. 3026.5
- c. 10165.4

Correct Answer: a. 9307.6

Q3: The missing factor in the equation $(P/A, i, n) * (---?---) = (A/P, i, n)$ is:

- a. $(P/G, i, n)$
- b. $(G/P, i, n)$
- c. $(A/P, i, n)$

Correct Answer: c. $(A/P, i, n)$

Q4: An interest rate of 8% per year compounded monthly is the same as:

- a. an effective 8.29% per year.
- b. an effective 8.29% per month.
- c. an effective 8% per year.

Correct Answer: a. an effective 8.29% per year.

Q5: The missing factor in the equation $(---?---) * (A/P, i, n) = (A/F, i, n)$ is:

- a. $(P/A, i, n)$
- b. $(P/F, i, n)$
- c. $(F/P, i, n)$

Correct Answer: c. $(F/P, i, n)$

Q6: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 8$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

- a. 6608.7
- b. 7527.2
- c. 1780.6

Correct Answer: a. 6608.7

Q7: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 6, we use:

- a. $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 6)$
- b. $7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 6)$
- c. $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 6)$

Correct Answer: c. $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 6)$

Q8: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 4, we use:

- a. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$
- b. $7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$
- c. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$

Correct Answer: a. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$

Q9: The missing factor in the equation $(P/F, i, n) * (---?---) = (A/F, i, n)$ is:

- a. $(P/A, i, n)$
- b. $(F/P, i, n)$
- c. $(A/P, i, n)$

Correct Answer: c. $(A/P, i, n)$

Q10: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. X(4) (equivalent single value at the end of year 4) is:

- a. -428.88
- b. -1713.3
- c. -583.49

Correct Answer: a. -428.88

Q11: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. X(7) (equivalent single value at the end of year 7) is:

- a. -1360.4
- b. -428.88
- c. -735.03

Correct Answer: b. -428.88

Q12: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount after 5 years is:

- a. JD 2250
- b. JD 75000
- c. JD 750

Correct Answer: c. JD 750

Q13: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 9$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

- a. 8427.6
- b. 7527.2
- c. 2180.8

Correct Answer: a. 8427.6

Q14: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 1, we use:

- a. $7000 + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$
- b. $7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 11)$
- c. $7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$

Correct Answer: c. $7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$

Q15: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(3)$ (equivalent single value at the end of year 3) is:

- a. -1850.9
- b. -540.27
- c. -428.88

Correct Answer: b. -540.27

Q16: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 7$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

- a. 1402.4
- b. 5675.1
- c. 6608.7

Correct Answer: c. 6608.7

Q17: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount after 11 years is:

- a. JD 3150
- b. JD 165000
- c. JD 1650

Correct Answer: c. JD 1650

Q18: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount after 3 years is:

- a. JD 1950
- b. JD 45000
- c. JD 450

Correct Answer: c. JD 450

Q19: The missing Factor in the equation $(P/A, i, n) * (F/P, i, n) = (---?---)$ is:

- a. $(F/A, i, n)$
- b. $(A/F, i, n)$
- c. $(P/F, i, n)$

Correct Answer: a. $(F/A, i, n)$

Q20: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 5, we use:

- a. $7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 7)$
- b. $7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 7)$
- c. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 7)$

Correct Answer: b. $7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 7)$